How MeSquare personalizes heart-health predictions

A plain-language guide to the personalization engine: what it predicts, how lab results and wearable data change the prediction, which machine learning and statistics it uses, and why. Written for readers without a statistics or data science background.

Model or rule	Version described
Population prior (ML)	20261104T1154Z
Activation registry	activation-20261104T1154Z
Risk equations	AHAprevent 1.1.0

Every number in the worked example (section 9) was produced by running the real engine on a sample user. Nothing in it is hand-made.

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1. The one-minute summary

MeSquare estimates five heart-health numbers for each user and turns them into a 10-year heart-disease risk. The five numbers are **non-HDL cholesterol**, **HDL cholesterol**, **HbA1c** (blood sugar over the past 3 months), **eGFR** (kidney function) and **systolic blood pressure**.

For each number the engine never gives one single value. It gives a **band**: a low, a typical and a high value. The band says “we are 80% sure the true value is in here.” A narrow band means we know a lot; a wide band means we are guessing.

The engine gets more personal in three layers:

Layer	When it is used	What kind of method	Shown to users today?
Population prior: what is typical for people like you	Always, from the first minute	Machine learning (LightGBM)	Yes
Lab anchor: your own lab result, with known lab noise	As soon as you upload a lab	Statistics (no learning)	Yes
Wearable adjustment: how your daily habits move the number	After 30+ days of wearable data	Machine learning (LightGBM)	Not yet: runs hidden (“shadow mode”) until proven

> **The golden rule: never worse.** A more personal layer is only shown when tests prove it beats the layer below it. Lab anchoring has passed. The wearable layer has not been tested on real people yet, so it stays hidden.

The final step sends the chosen numbers to **AHA PREVENT**, the official American Heart Association risk calculator, run exactly as the AHA publishes it.

2. Words you will see

Only the terms needed to read the rest. Section 14 has the full glossary.

Term	Plain meaning
Biomarker	A measurable health number, such as HDL cholesterol or blood pressure.
Prior	Our best guess before we know anything personal. Here: what is typical for people with your age, sex, weight and history.
Band / interval	A range (low to high) the true value probably sits in.
p10, p50, p90	The low, middle and high ends of the band. p10 = only 10% of people are below it; p50 = the typical value (median); p90 = only 10% are above it. p10 to p90 holds 80% of cases.
Coverage	How often the truth really lands inside the band. An honest 80% band should contain the truth about 80% of the time.
Lab anchor	Using your own most recent lab result as the centre of your band, instead of the population guess.

Term	Plain meaning
CV (coefficient of variation)	Noise as a percentage. A CV of 6.5% means repeat measurements typically differ by about 6.5%.
Shadow mode	A model runs and its answers are saved, but users never see them. Used to test a model safely.
Machine learning (ML)	A computer finds patterns in many examples, instead of people writing the rules by hand.

3. The big picture: one request, eight steps

Every time the app needs a fresh picture of a user, the backend sends the user's profile and *every* lab and wearable record to the engine. The engine keeps no memory between requests: it rebuilds the whole picture each time. That makes it easy to reproduce any answer later.

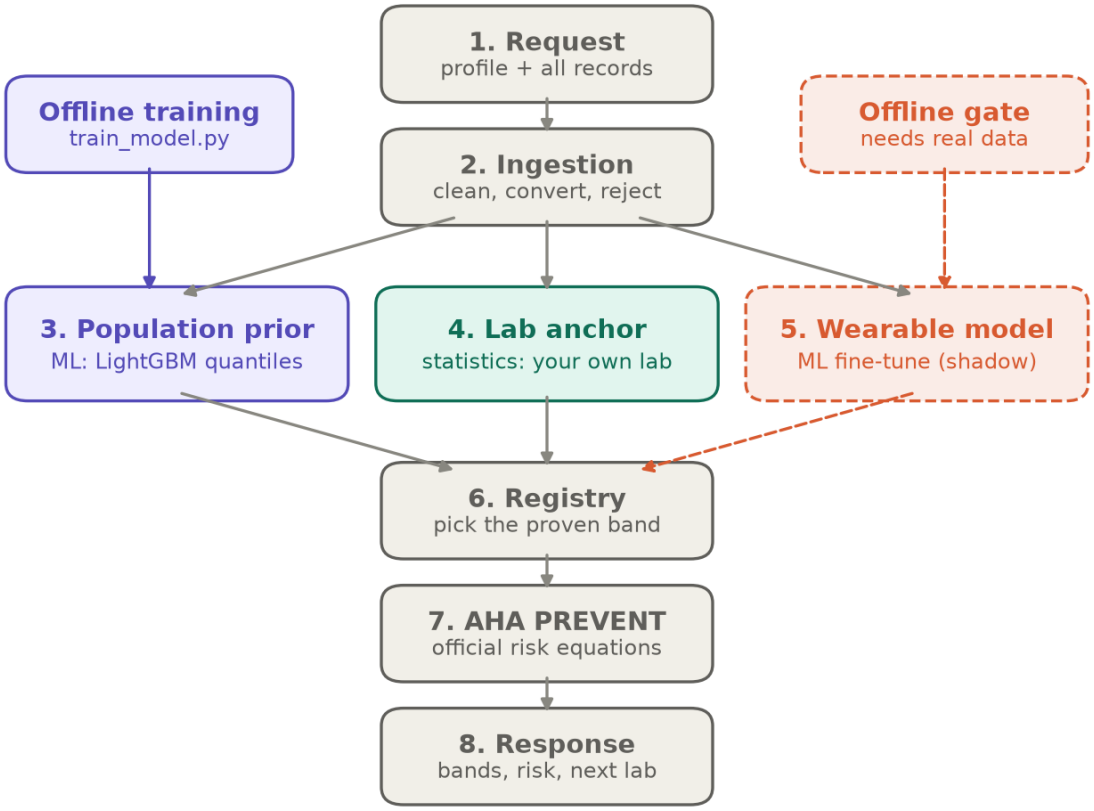


Figure 1. One request through the engine. Purple = machine learning, green = statistics, orange dashed = machine learning that is still hidden (shadow mode), grey = fixed rules.

Step	What happens	Type
1. Request	Profile (age, sex, height, weight, smoking, medications, diabetes) and all records arrive.	Input

Step	What happens	Type
2. Ingestion	Each record is checked: name recognised, unit converted (e.g. mmol/L to mg/dL), value plausible, method is a real lab test. Bad records are rejected with a reason. Missing pieces are derived (non-HDL = total minus HDL; eGFR from creatinine).	Rules
3. Population prior	The trained ML model gives a band for each biomarker from the profile alone.	ML
4. Lab anchor	If there is a usable lab, a tighter band is built around it.	Statistics
5. Wearable model	Wearable trends are summarised; the fine-tuning model's answer is saved, hidden.	ML (hidden)
6. Registry	A settings file says which layer is proven for each biomarker; the best proven one is shown.	Rules
7. AHA PREVENT	The chosen values go into the official risk equations.	Rules (AHA)
8. Response	Bands, risk, a “get your next lab” reminder, model versions and a fingerprint of the input.	Output

4. Step A: the population prior (machine learning)

What it answers

“Before we have any lab from you: where do people like you usually sit?” A 52-year-old man who weighs 92 kg does not have the same typical cholesterol as a 30-year-old woman. The prior captures those differences.

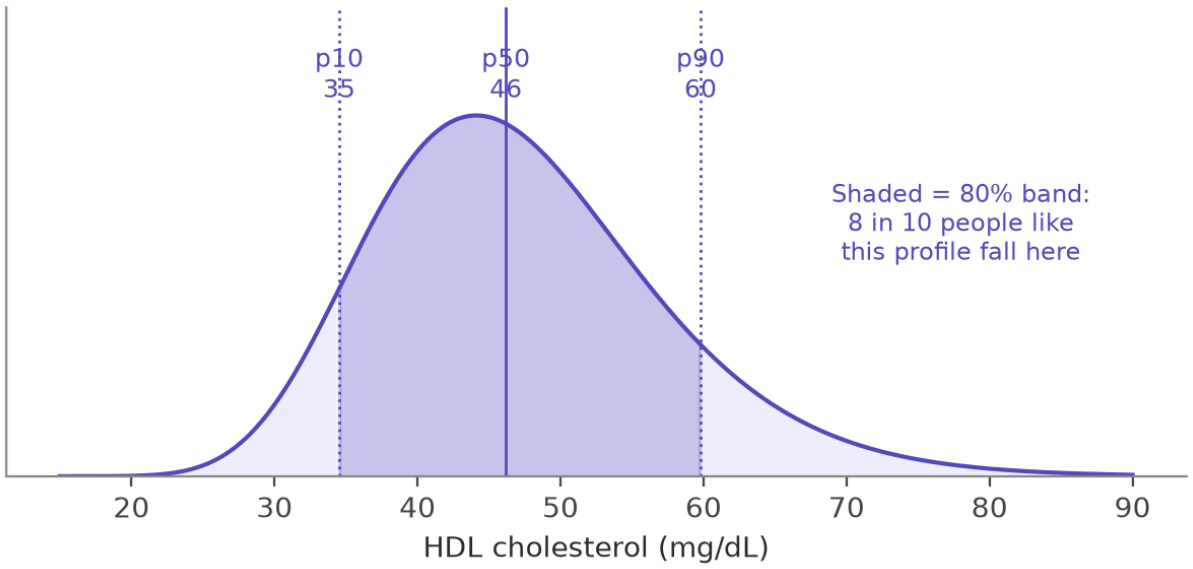


Figure 2. The population band for HDL cholesterol of our sample user (52-year-old man, 92 kg, no labs). The typical value is 46, but 80% of similar people fall anywhere from 35 to 60.

Where the knowledge comes from

The model learned from **NHANES 2021-2023**, a US government health survey where thousands of people were examined and had blood drawn. Each person in NHANES carries a **survey weight** (how many Americans they represent), and the model uses those weights so the results reflect the US population, not just the people who happened to be sampled.

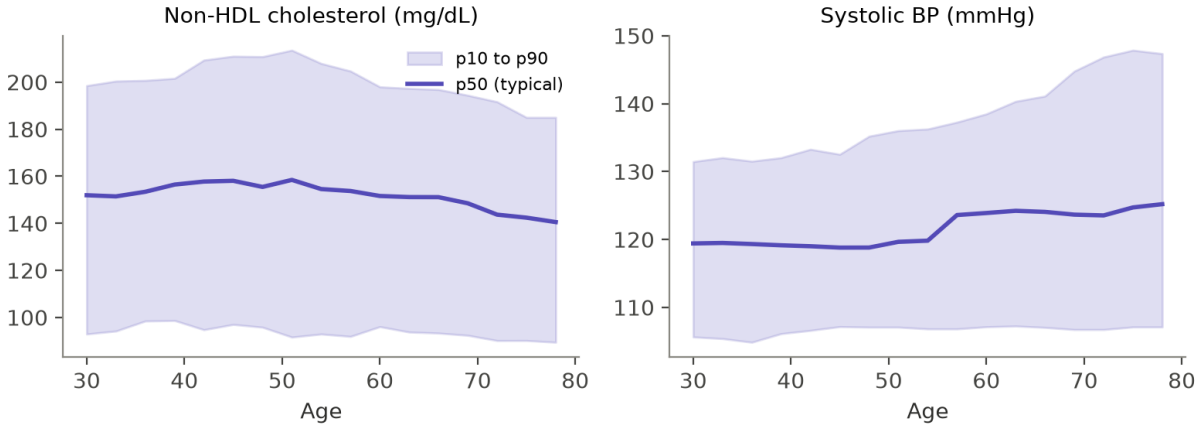


Figure 3. What the trained model has learned: the typical value (line) and 80% band (shaded) change with age, for the same man at different ages. These curves come from the real model.

The algorithm: LightGBM quantile regression

What it is. LightGBM builds hundreds of small decision trees. A decision tree is a flowchart of yes/no questions ("Is age over 50? Is BMI over 30?") ending in a number. One tree is a rough guess; each next tree learns to fix the mistakes left by the trees before it. This is called **gradient boosting**.



Each small tree corrects the leftover errors of the trees before it (gradient boosting).
One set is trained for p10, one for p50, one for p90 (quantile loss).

Figure 4. Gradient boosting in one picture.

Quantile regression means that instead of training one model to predict the average, we train three: one for p10, one for p50 and one for p90. Each uses a scoring rule (**pinball loss**) that rewards it for having exactly 10%, 50% or 90% of people below its line. Together they draw the band.

Why this algorithm.

- It handles mixed inputs (numbers like age, categories like smoking) and missing values without extra work.
- It finds non-straight-line patterns, such as risk rising faster after 60.
- It is fast: a prediction takes milliseconds, so the app stays responsive.
- It gives a *band*, not just one number, which is what personalization needs.
- It is well understood and tested in medical research, which matters for audits.

Making the band honest: calibration

A model can claim an 80% band but really be right only 70% of the time. We test and fix that in three ways:

- **Grouped cross-validation.** The data is split into parts; the model trains on some parts and is tested on a part it has never seen. NHANES samples people in geographic clusters, so whole clusters are kept together, otherwise neighbours would "leak" answers into the test.
- **Conformal widening.** Using those test results, each band is stretched just enough that it really covers 80%. This is done separately for men and women and for three age groups (20-39, 40-59, 60+), so no group is short-changed. (Called "Mondrian conformal" calibration.)
- **Outlier rule.** Values that are physically implausible are removed before training, using the same limits the app uses to reject bad uploads.

> **Result:** on people the model never saw, the 80% bands contained the truth 79-82% of the time, and no sex or age group was off by more than 0.7 percentage points.

Training happens once, offline (`train_model.py`). Each user's request only *uses* the trained model. Nothing is retrained per user.

5. Step B: the lab anchor (statistics)

What is a lab anchor?

Once a user uploads a real lab result, we stop relying on "people like you" and centre the band on **the user's own value**. The lab is the anchor: the band is tied to it, like a boat to an anchor, and can only drift a short way from it.

We do **not** take the lab as perfectly exact. Every lab result has two kinds of noise:

- **Analytical noise**: the machine itself varies a little between runs.
- **Biological noise**: your body varies day to day (what you ate, sleep, stress, hydration).

Medical research has measured both for each biomarker (the European EFLM Biological Variation Database). We store them as a single percentage per biomarker:

Biomarker	Single-measurement noise (CV)	Slow drift over a year
Non-HDL cholesterol	8.5%	8%
HDL cholesterol	6.5%	6%
HbA1c	1.3%	4%
eGFR	6.0%	4%
Systolic BP	6.0%	5%

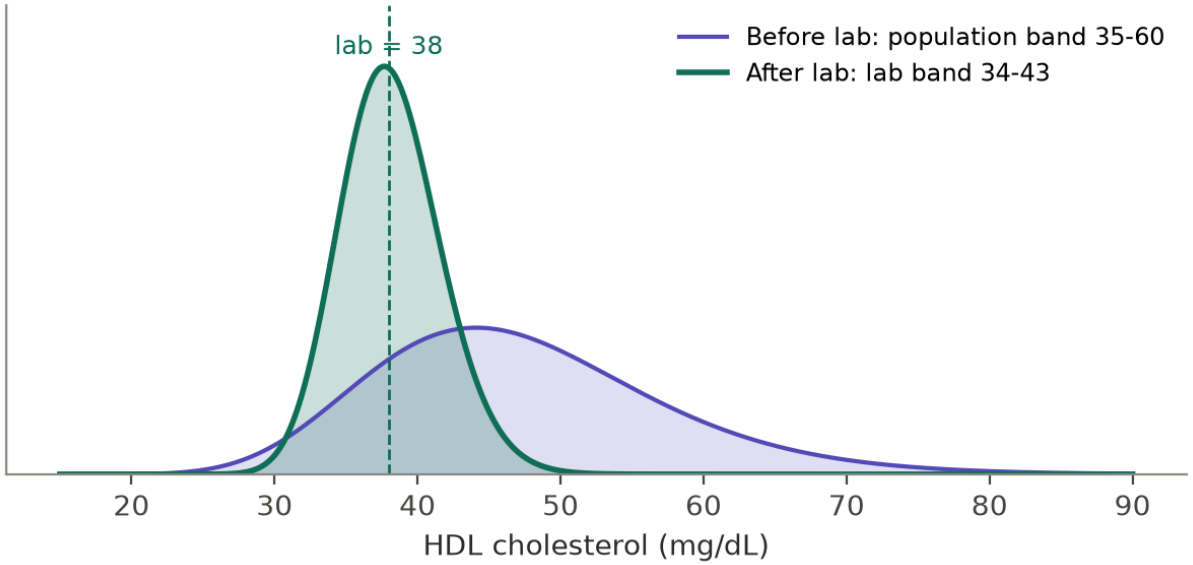


Figure 5. What happens when the first lab arrives. The user's HDL came back at 38, lower than typical. The band jumps to centre on 38 and becomes much narrower: we now know far more.

How the band is computed (gently)

Three ingredients decide how wide the band is around the lab value:

Ingredient	Why it is there
Noise of the old lab	The lab we anchor on was itself a bit off.
Noise of the next lab	We predict what your *next* test will show, and it will be a bit off too.
Drift since the lab	Your true level slowly wanders. The older the lab, the more it may have moved. Drift grows with the square root of time, so it grows fast at first, then slower.

For readers who want the formula, the spread on a log (percentage) scale is:

$\text{spread} = \sqrt{2 \times \text{lab noise}^2 + \text{drift}^2}$ and band = lab $\times e^{\pm 1.28 \times \text{spread}}$

The 1.28 is the number that turns a spread into an 80% band. Percentages are used because noise scales with the value: a 10% wobble on 200 is bigger than on 40. This also keeps the band from ever going below zero.

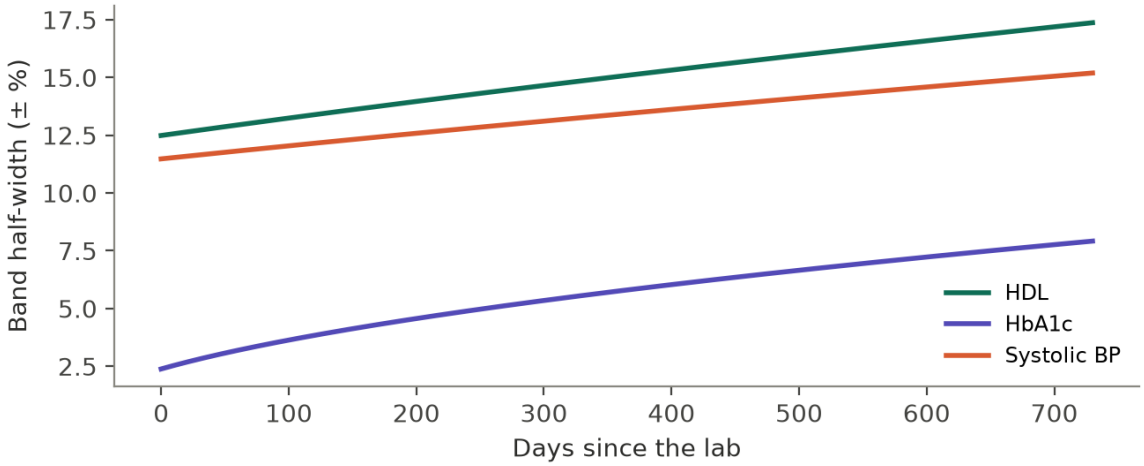


Figure 6. The band slowly widens as a lab gets older. HbA1c is measured very precisely, so its band stays tight; blood pressure is noisier.

Why statistics here, not machine learning?

- It works from the very first lab. No training data from our users is needed.
- Every number comes from published medical research, so it is easy to explain and audit.
- We tested it (section 11): the error was cut by 26-51% compared with the population guess.

> **Why not blend the lab with the population guess?** An earlier design used a Kalman filter to average the two. A single lab carries far more information than the population band, so the blend ends up almost exactly the lab anyway, and the filter needed extra assumptions (such as how much each habit moves each biomarker) that we cannot check without real repeat data. The simpler lab anchor performed as well and is easier to explain.

When the anchor is dropped (resets)

Some events make an old lab no longer describe the user. The engine then ignores labs from before the event and falls back to the population band until a new lab arrives:

Event	Labs ignored
Started a statin (cholesterol drug)	Cholesterol labs before the start date
Started blood-pressure medication	Blood-pressure readings before the start date
Pregnancy	All labs

5b. The lab anchor step by step

This part opens up the lab anchor completely: where its noise numbers (CVs) come from, every formula it uses, why each formula was chosen, and the exact order the engine applies them. One real case runs through it all: the sample user's **HDL of 38 mg/dL**, measured on day 14.

Where the CV numbers come from

CV (coefficient of variation) is the size of the noise as a percentage of the value. Measure the same thing many times; the CV is how much the results typically differ from their average, divided by that average:

```
CV = SD / mean
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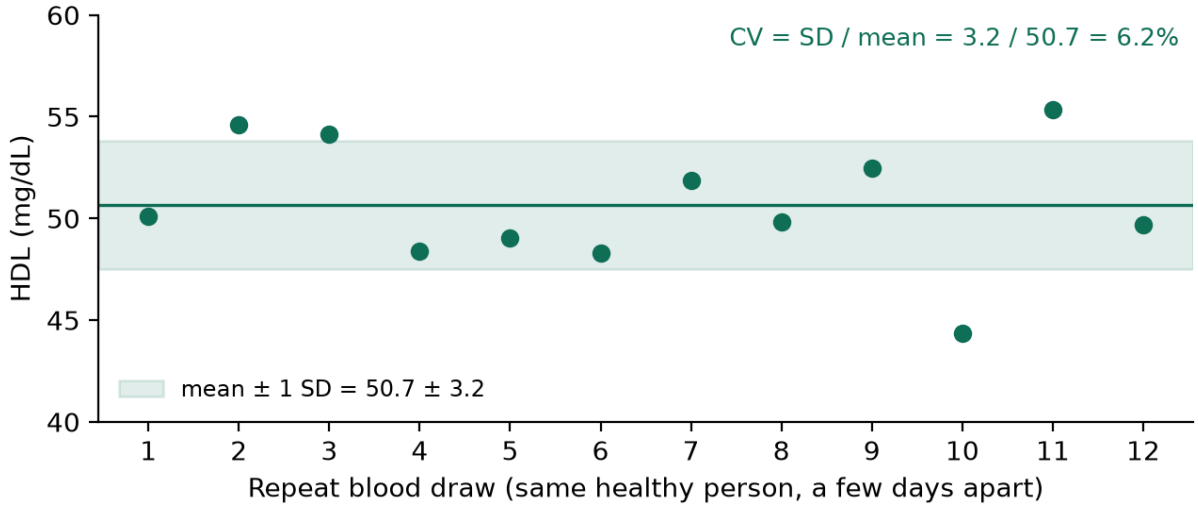


Figure 7. How a CV is measured: draw blood from the same healthy, stable person many times. The results scatter even though the person did not change. That scatter, as a percentage of the mean, is the CV. (Illustration with simulated draws.)

A single lab result is affected by two separate kinds of noise, each with its own CV:

Part	What causes it	How it is measured
Analytical CV (CV_A)	The lab machine and chemistry: two runs of the same tube differ a little.	The lab runs the same control sample again and again (quality control) and computes SD / mean.
Within-person biological CV (CV_I)	The body itself: meals, sleep, hydration, stress, time of day.	Research studies take repeated samples from healthy people over weeks, then separate machine noise from body noise with a statistical method called nested ANOVA.

The two are independent, so they combine by adding their squares (the same rule used throughout this section):

$$CV_{total} = \sqrt{(CV_A^2 + CV_I^2)}$$

Worked example for HDL: the EFLM database gives CV_I = 5.7% [7]. The EFLM “desirable” analytical imprecision is at most half of that, CV_A = 2.85% [23]. Combined: $\sqrt{(5.7^2 + 2.85^2)}$ = 6.4%. The engine uses 6.5%.

Where our values come from. We do not calculate CVs from our own users. We take CV_I from the **EFLM Biological Variation Database** (the European Federation of Clinical Chemistry and Laboratory Medicine's curated meta-analysis of these studies) and CV_A from the EFLM desirable specification ($0.5 \times CV_I$). Section 13.3 shows the calculation and source for every biomarker. The values are kept in one file, `biomarker_config.py`:

Biomarker	measurement_cv (one lab)	annual_drift_cv (per year)
Non-HDL cholesterol	8.5%	8%
HDL cholesterol	6.5%	6%
HbA1c	1.3%	4%
eGFR (from creatinine)	6.0%	4%
Systolic BP (home protocol)	6.0%	5%

Section 13 lists the exact source of each number, and which ones (the drift values) are our own estimates rather than published.

A new user uploads their first lab: how is the CV calculated? It is not. With one value there is no SD or mean to compute. The CV is a fixed percentage per biomarker, measured once by research on many other stable people, and the same for every user. The engine applies it to this user's value: 6.5% of an HDL of 38 is about 2.5 mg/dL of expected noise. The user's own value is the centre; the noise size is borrowed from published research. A personal CV would need several repeat labs taken while nothing changed, which most users will never have.

Why published values instead of our own? We cannot estimate within-person noise without many people who have repeated labs close together while nothing changed. We do not have that data yet; published studies do, collected under controlled conditions. Once enough of our users have repeat labs, the same CV = SD / mean idea can be used to check, and if needed adjust, these numbers.

Every formula, and why it was chosen

#	Formula	What it does	Why this one
F1	$CV = SD / mean$	Defines noise as a percentage.	Lab noise grows with the value, so a percentage describes it at every level.
F2	$CV_{total} = \sqrt{(CV_A^2 + CV_I^2)}$	Combines machine and body noise.	Independent noises add by their squares (variances), not directly.
F3	$\sigma = \sqrt{\ln(1 + CV^2)}$	Turns a CV into a spread on the log (percentage) scale.	Exact conversion for a log-normal value. Log-normal fits skewed biomarkers and never goes below zero.
F4	$\sigma_{drift} = \sigma_{year} \times \sqrt{(days / 365)}$	Spread from slow natural change since the lab.	Random-walk model: many small independent nudges, so variance grows in step with time.

#	Formula	What it does	Why this one
F5	$\sigma = \sqrt{(2 \cdot \sigma_{meas}^2 + \sigma_{drift}^2)}$	Total spread of the next lab around this lab.	Two noisy labs (this one and the next) plus drift, all independent, so variances add.
F6	$p10, p90 = lab \times e^{\mp 1.2816 \cdot \sigma}$	Turns the spread into the 80% band.	1.2816 leaves 10% below and 10% above, matching the population model's p10 to p90.
F7	change = new / old − 1	Percentage change between two labs.	Same percentage scale as the noise.
F8	$limit = 1.96 \times \sqrt{2} \times \sigma_{meas}$; $RCV = e^{\pm limit} - 1$	Largest change noise alone explains (95%).	Standard Reference Change Value, log-normal form (Fokkema 2006): √2 because both labs are noisy, 1.96 for 95%.

The engine's steps, in order

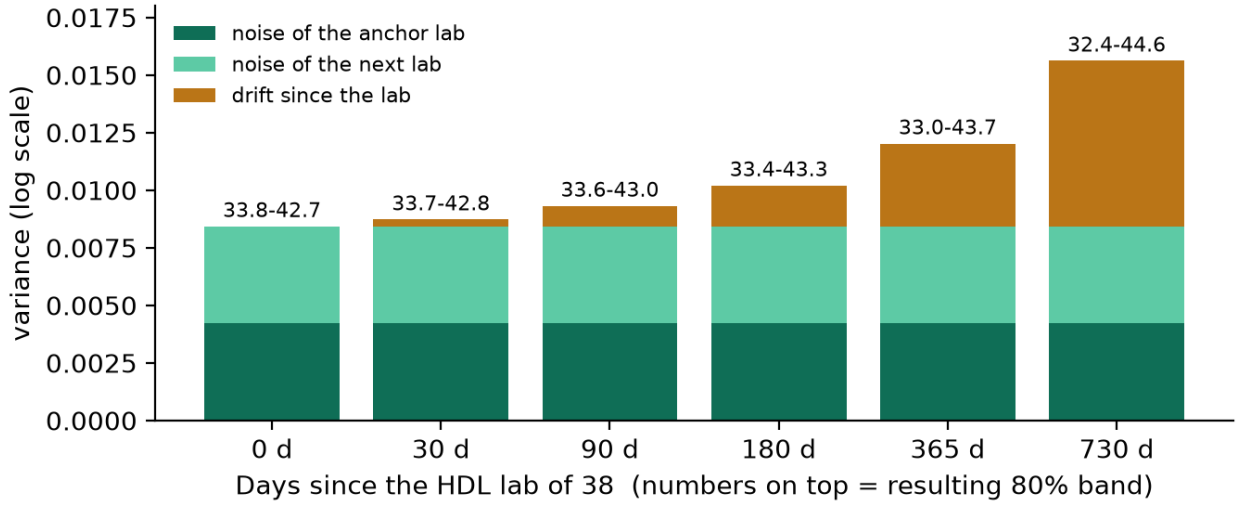


Figure 8. What makes up the band for the HDL lab of 38. The two lab-noise pieces never shrink; drift grows with time. Numbers on top are the resulting 80% bands from the engine's formulas.

Step	What the engine does	Why	HDL example
1. Keep only labs that still describe the user	Measured lab methods only (lab report, clinic, home BP protocol), taken on or before today, and not cancelled by a reset (statin for cholesterol, BP medication for blood pressure, pregnancy for all).	An anchor is only useful if it still describes the person. With none left, the biomarker goes back to the population band.	HDL 38 from day 14 is kept.

Step	What the engine does	Why	HDL example
2. Take the latest lab as the centre (p50)	The newest usable lab becomes the middle of the band.	It is the closest to the person's level today. Older labs are not averaged in, because the person may have changed; they are used only for the change test.	p50 = 38.
3. Convert the noise CV into a log-scale spread (F3)	$\sigma_{meas} = \sqrt{\ln(1 + 0.065^2)}$	Percent-based noise needs the log scale; this is the exact conversion.	$\sigma_{meas} = 0.0649$
4. Count the lab noise twice (part of F5)	$2 \times 0.0649^2 = 0.0084$	We predict the next lab: the old lab was a bit off, and the next one will be too.	Lab-noise variance = 0.0084
5. Add drift for the days since the lab (F4)	$\sigma_{drift} = \sqrt{\ln(1 + 0.06^2)} \times \sqrt{(\text{days} / 365)}$	The true level slowly wanders; an older lab is less certain.	0 days: 0; 90 days: 0.0298
6. Combine into one spread (F5)	$\sigma = \sqrt{(0.0084 + \sigma_{drift}^2)}$	Independent noises add as variances.	0 days: 0.0918; 90 days: 0.0965
7. Build the 80% band (F6)	$38 \times e^{\pm 1.2816 \times \sigma}$	Gives p10 and p90 on the same 80% footing as the population band.	Day 0: **33.8 to 42.7**; 90 days later: **33.6 to 43.0** (population band was 34.5 to 59.8).
8. Check it may be shown	Clamp to the plausible range; the activation registry must have the lab stage on for this biomarker (it is on for all five).	Safety net against impossible values, and the never-worse rule.	Basis = lab.
9. With 2+ labs: real-change test (F7, F8)	Compare the latest lab with the previous one against the RCV.	Separates real change from noise, with about 5% false alarms by design.	HDL 38 to 43 = +13.2%, inside −16.5% to +19.7%: within noise. Non-HDL 192 to 150 = −21.9%, beyond −21.0%: real change.
10. Hand the value to the risk calculator	PREVENT receives the measured lab itself (not the band), marked “measured”.	A measured value is the best input; it also removes that input from the risk range.	PREVENT uses HDL = 38.

In one sentence: the lab anchor says “your next result will most likely be your last result, give or take the known noise of two lab tests plus the natural drift since then”, and every piece of that sentence is one of the formulas above, with numbers from published laboratory research.

6. Step C: repeat labs and the real-change test

With two or more labs, the newest becomes the anchor, and the engine also answers a question users care about: **“Did my number really change, or is this just noise?”**

It uses the **Reference Change Value (RCV)**, a standard laboratory-medicine test. From the known noise of two measurements, it computes how big a change could happen by chance alone 95% of the time. Only a bigger change is flagged as real.

Biomarker	Largest fall noise can explain	Largest rise noise can explain
Non-HDL cholesterol	-21.0%	+26.5%
HDL cholesterol	-16.5%	+19.7%
HbA1c	-3.5%	+3.7%
eGFR / Systolic BP	-15.3%	+18.1%

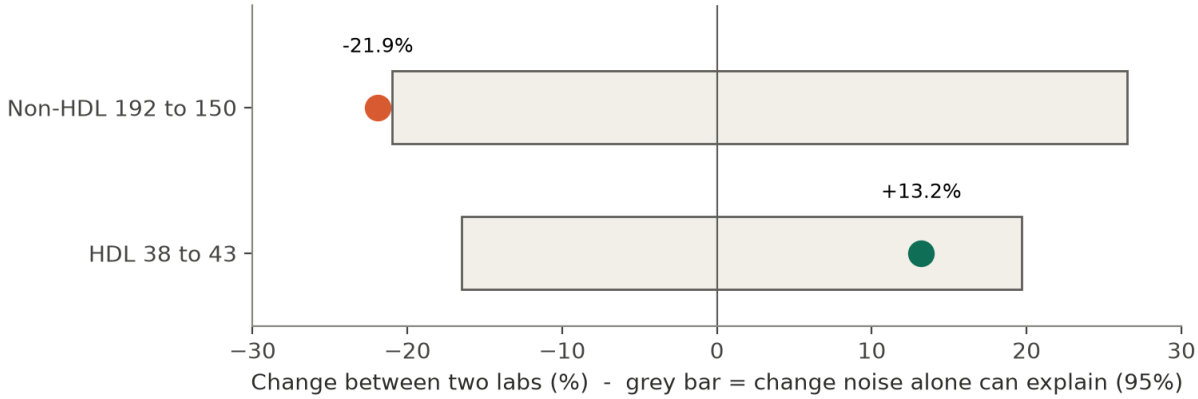


Figure 9. The sample user's second lab. Non-HDL fell 21.9%, beyond the noise zone: flagged as a real change. HDL rose 13.2%, which is encouraging but still inside the noise zone, so it is not flagged yet.

Falls and rises have different limits because noise works in percentages: dropping from 100 to 85 and rising from 85 to 100 are the same step on a percentage scale. (This form of the test is from Fokkema et al., Clinical Chemistry, 2006.)

> By design, this test gives a false alarm about 5% of the time when nothing really changed. Our simulation measured 4.3-5.7%, matching the design.

7. Step D: wearable data and the fine-tuning model

How wearable data comes in

The backend collects daily summaries from devices (via Open Wearables) once a day and sends them with every request. The engine uses four daily metrics: **resting heart rate**, **heart-rate variability (HRV)**, **VO2 max** (fitness) and **weight** (from a smart scale). Each record must come from a wearable or smart-scale method; anything else is rejected.

Turning daily data into trends (windows)

Single days are noisy, so the engine summarises each metric over the last **30** and **90** days. For each window it computes:

Summary	Plain meaning
Mean	Your usual level over the window.
SD (standard deviation)	How much it bounced around day to day.
Slope (Theil-Sen)	Whether it is trending up or down, per day. Theil-Sen is a robust trend line: one strange day cannot drag it around.
Coverage	The share of days in the window that actually have data.

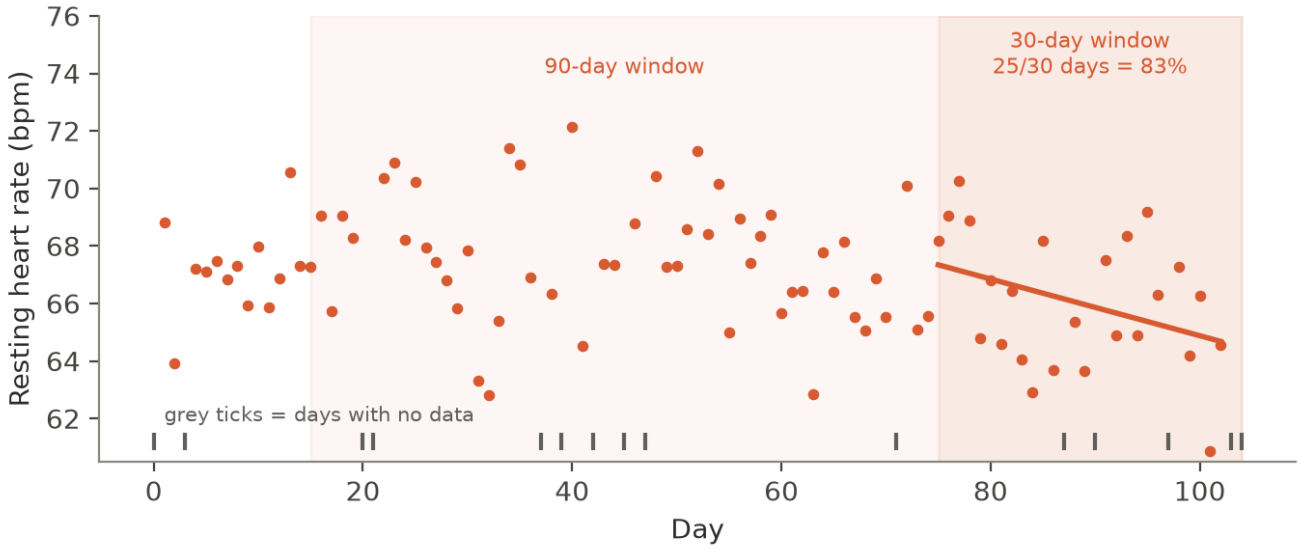


Figure 10. The sample user's resting heart rate. Grey ticks are days the watch was not worn. The 30-day window still has 83% coverage, so it is used; its trend line shows a gentle fall.

Missing days

People forget to wear devices or turn off tracking. That is expected and handled simply: a window is used only if **at least 70%** of its days have data. If not, that window is marked "insufficient" and left out; the rest of the engine carries on with the lab or population band. Missing days are never filled in with made-up values.

The fine-tuning model: what it does

We do not retrain the big population model for each user. Instead, a second, small LightGBM model learns a **correction** (called a **residual**): "given how this person's heart rate, HRV, fitness and weight have been trending, how far will their next lab land from our current band centre?" It

has two modes:

Mode	For which users	Corrects
Level	Wearables but no lab yet	The population typical value
Change	Has a lab, plus 60+ days of wearables since it	The last lab value

The correction is a percentage shift plus its own uncertainty band, also calibrated with conformal widening.

Shadow mode and the gate

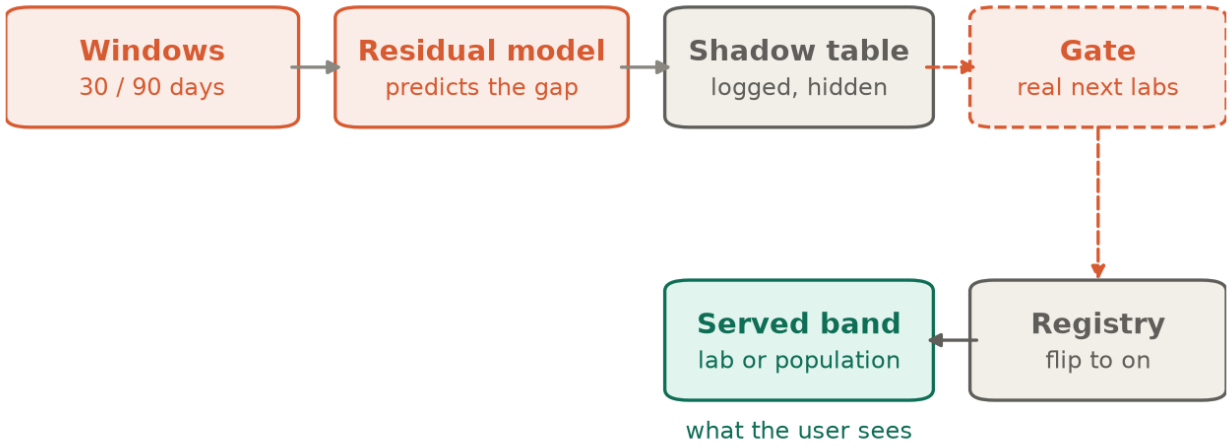


Figure 11. The wearable model's answers are saved to a hidden table. Users keep seeing the lab or population band until the gate is passed.

To be switched on for a biomarker, the model must pass **all** of these checks on real, consented users whose next lab later arrived:

- At least **200 users** with wearable data followed by a lab.
- At least **10% more accurate** than both the last lab alone and the population band (tested on users it never trained on).
- The improvement holds up when we re-sample users 2,000 times (the lower end of the confidence interval stays above zero).
- Its 80% band really contains the next lab 75-85% of the time.

It is switched on in a fixed order: HbA1c first, then HDL and non-HDL. eGFR always stays lab-only, because kidney function does not respond to daily habits on these time scales.

Today: there are no real paired data yet, so no wearable model has been trained. The engine already computes and returns the wearable windows, and as soon as a trained model is placed in the model folder it runs in shadow mode automatically. Users see no wearable-driven change until the gate passes.

8. Step E: choosing what to show, and the heart-risk score

The activation registry

A small settings file (`config/activation.json`) records, for each biomarker and each layer, whether it is switched on, and the evidence behind that decision. The engine shows the most personal layer that is switched on:

Situation for one biomarker	What the user sees	Label (basis)
No usable lab	Population band	population
Usable lab (no reset since)	Lab band	lab
Lab + wearable layer proven	Lab band, adjusted by wearables	lab_wearable
No lab + wearable layer proven	Population band, adjusted	population_wearable

AHA PREVENT: from five numbers to a risk

PREVENT (Predicting Risk of cardiovascular disease EVENTs) is the American Heart Association's 2023 risk equation. We run the official AHA R package unchanged, so the risk math is exactly the AHA's, not our own copy. It needs total and HDL cholesterol, blood pressure, BMI, eGFR, diabetes, smoking and medications.

- For each biomarker the engine passes the **measured lab** if there is one, otherwise the **predicted typical value** (p50).
- The risk range comes from running PREVENT again with the band's low and high values (the combinations that push risk lowest and highest). A measured input has no range, so with all labs measured the range collapses to one number.
- Category for 10-year total heart disease risk: **low** under 5%, **borderline** 5-7.5%, **intermediate** 7.5-20%, **high** 20% or more.
- **risk_basis** tells the user how solid the number is: *measured* (all labs), *mixed*, or *estimated* (no labs).
- PREVENT is validated only for ages 30-79. Outside that, risk is not generated; inputs outside the AHA's tested ranges are clipped and flagged.
- Risk values are percentages: 5.3 means 5.3%, not 530%.

9. End-to-end example: one user over 230 days

Meet our sample user: a **52-year-old man**, 178 cm, 92 kg, never smoked, no diabetes, no medications. He connects a watch and smart scale on day 0 and wears them about 4 days out of 5. Every number below came from running the real engine.

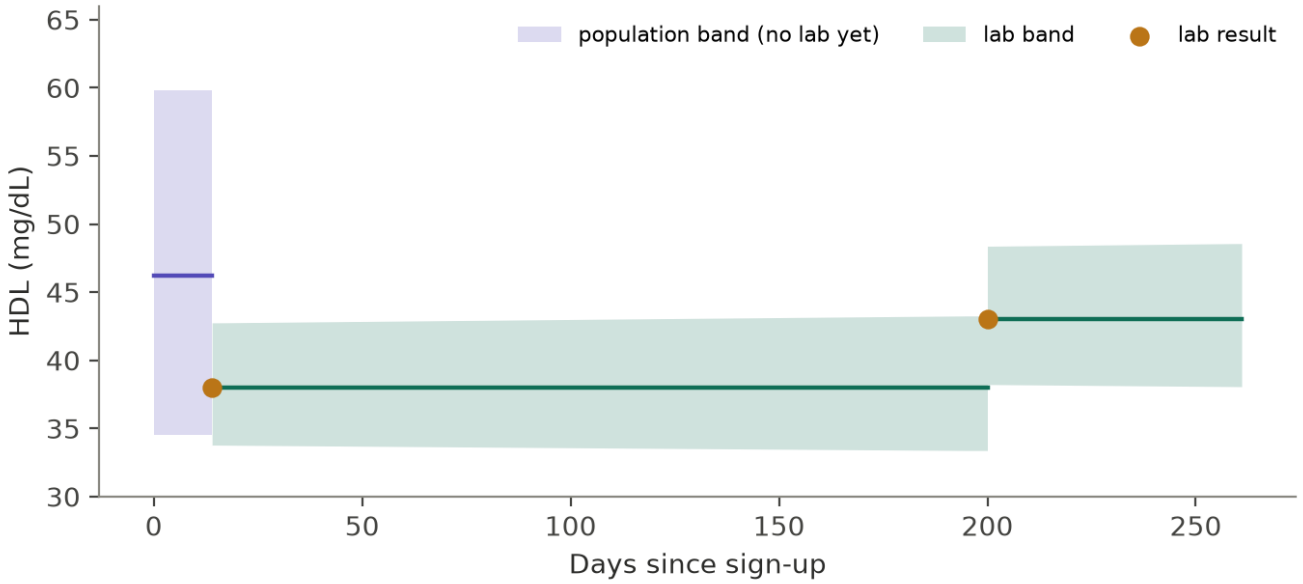


Figure 12. His HDL band over time: wide population band at sign-up, a tight band anchored to his first lab on day 14, slowly widening, then re-anchored by his second lab on day 200.

Day 0: signs up, no labs

With no labs, every band is the population band. The risk is an estimate with a wide range. The app asks for a first lab right away.

Biomarker	Band: typical (low to high)	Shown from
Non-HDL chol. (mg/dL)	155 (92 to 212)	population
HDL chol. (mg/dL)	46 (35 to 60)	population
HbA1c (%)	5.8 (5.3 to 6.6)	population
eGFR (kidney)	101 (81 to 109)	population
Systolic BP (mmHg)	120 (107 to 137)	population

10-year heart-disease risk: 3.3% (range 2.1 to 12.2), low; basis *estimated*. **Next lab:** due now.

Day 14: uploads first lab panel

He uploads total cholesterol 230, HDL 38, HbA1c 5.9%, creatinine 1.0 and a home blood-pressure reading of 138. The engine derives non-HDL (230 - 38 = 192) and eGFR from creatinine. Every band snaps to his own values and narrows. His risk is now based on measured values, so it has no range, and it moves up into borderline because his real numbers are worse than typical.

Biomarker	Band: typical (low to high)	Shown from
Non-HDL chol. (mg/dL)	192 (165 to 224)	lab
HDL chol. (mg/dL)	38 (34 to 43)	lab
HbA1c (%)	5.9 (5.8 to 6.0)	lab
eGFR (kidney)	91 (81 to 101)	lab
Systolic BP (mmHg)	138 (124 to 154)	lab

10-year heart-disease risk: 5.3%, borderline; basis *measured*. **Next lab:** due 2026-04-19.

Day 104: no new lab, 90 days later

Nothing new from the lab. The bands are the same centre but very slightly wider (drift). The lab is 90 days old, so the app now suggests a follow-up lab. Wearable windows are computed (resting heart rate 30-day coverage 83%, trending down), but they do not change what he sees.

Biomarker	Band: typical (low to high)	Shown from
Non-HDL chol. (mg/dL)	192 (163 to 226)	lab
HDL chol. (mg/dL)	38 (34 to 43)	lab
HbA1c (%)	5.9 (5.7 to 6.1)	lab
eGFR (kidney)	91 (81 to 101)	lab
Systolic BP (mmHg)	138 (123 to 155)	lab

10-year heart-disease risk: 5.3%, borderline; basis *measured*. **Next lab:** due now.

Day 200: second lab after lifestyle changes

New panel: total cholesterol 193, HDL 43, HbA1c 5.6%, blood pressure 128. Non-HDL fell from 192 to 150 (-21.9%): flagged as a real change, and so is HbA1c 5.9% to 5.6% (-5.1%), because HbA1c is measured very precisely. The HDL and blood-pressure improvements are inside the noise zone, so not flagged yet. Risk drops back to low.

Biomarker	Band: typical (low to high)	Shown from
Non-HDL chol. (mg/dL)	150 (129 to 175) change -21.9%: real change	lab
HDL chol. (mg/dL)	43 (38 to 48) change +13.2%: within noise	lab
HbA1c (%)	5.6 (5.5 to 5.7) change -5.1%: real change	lab
eGFR (kidney)	91 (81 to 101) change +0.0%: within noise	lab
Systolic BP (mmHg)	128 (115 to 143) change -7.2%: within noise	lab

10-year heart-disease risk: 3.9%, low; basis *measured*. **Next lab:** due 2026-10-22.

Day 230: doctor starts a statin on day 225

A statin changes cholesterol, so his old cholesterol labs no longer describe him. Cholesterol falls back to the population band for a man on a statin, and the risk becomes mixed (partly measured, partly estimated) with a range. HbA1c, eGFR and blood pressure keep their lab anchors. A new cholesterol lab will re-anchor them.

Biomarker	Band: typical (low to high)	Shown from
Non-HDL chol. (mg/dL)	120 (67 to 197)	population
HDL chol. (mg/dL)	46 (34 to 59)	population
HbA1c (%)	5.6 (5.4 to 5.8) change -5.1%: real change	lab
eGFR (kidney)	91 (81 to 101) change +0.0%: within noise	lab
Systolic BP (mmHg)	128 (115 to 143) change -7.2%: within noise	lab

10-year heart-disease risk: 3.0% (range 2.0 to 5.1), low; basis *mixed*. **Next lab:** due 2026-10-22.

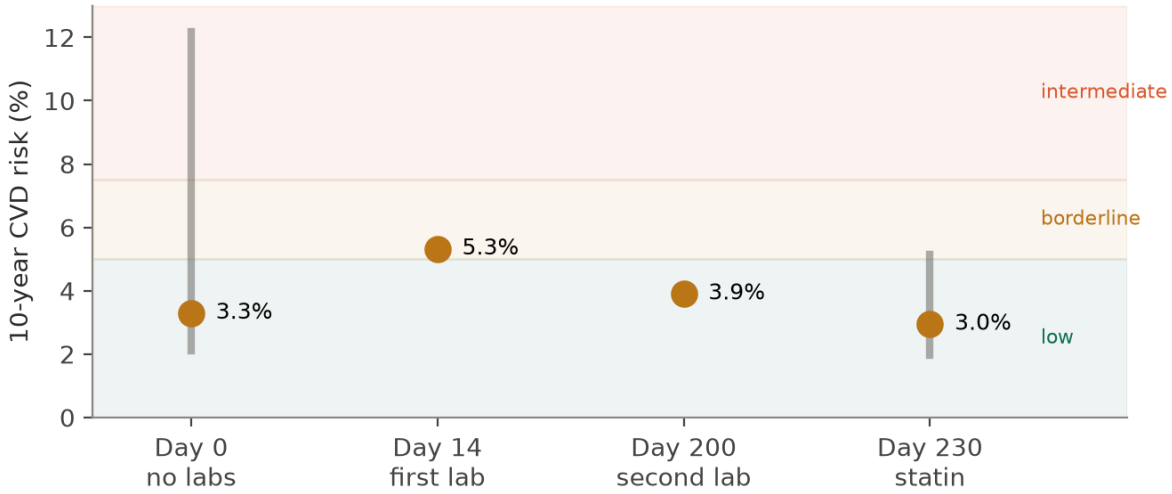


Figure 13. His 10-year risk at each moment. The dot is the best estimate; the grey bar is the range, which disappears when every input is measured.

9b. Onboarding scenarios A-F: what the backend sends and gets back

Six typical new users, run through the real engine on 4 November 2026. Every user is the same 52-year-old man (178 cm, 92 kg, never smoked, no medications); only what he uploads differs. Every number below is the engine's actual response.

Scenario	Labs	Wearables
A	none	none
B	full panel, 2 days ago	none
C	full panel, 45 days ago	none
D	none	20 days (under 30)
E	none	45 days (over 30)
F	full panel 75 days ago, before wearables	70 days, after the lab

The one endpoint the backend calls

Every scenario uses POST `/v1/twin/state` (`app.py: twin_state_endpoint`). The engine keeps no memory, so each call sends the profile and **every** record the user has. Call it after onboarding, after every lab upload, once a day after the end-of-day wearable sync, and whenever the profile changes (for example a statin start). `/v1/predict` returns the population prior only and is not meant for the app.

```
{"profile": {"sex": "Male", "age": 52, "race_eth": "Non-Hispanic White", "height_cm": 178, "weight_kg":
92, "smoking": "never", "bp_meds": 0, "chol_meds": 0, "diabetes_dx": "no", "statin_started_at": null,
"bp_meds_started_at": null, "pregnant": false},
 "records": [{"id": "...", "biomarker_abbr": "HDL", "value": 38, "unit": "mg/dL", "measurement_method":
"blood_test_lab_report_upload", "measured_at": "2026-11-02T08:00:00Z"}, ...],
 "as_of": "2026-11-04", "user_id": "user-123"}
```

The response has: `predictions` (one per biomarker: basis, basis_label, p10/p50/p90, source_lab_id, lab_measured_at, change), `risk`, `rejected_records`, `wearable_windows`, `lab_prompt`, `model_versions` and `input_hash`.

Which code runs at each step, and why

All steps run inside `twin_state()` in `src/pipeline/twin_pipeline.py`, in this order:

Step	Code	What it does	Why
1. Ingest records	record_ingestion.py: ingest_records() → parse_record(), convert_unit(), derive_values()	Recognise names, convert units, reject implausible or non-lab values, derive non-HDL and eGFR.	Bad input must never reach the models; rejections are reported with a reason instead of failing the call.

Step	Code	What it does	Why
2. Population prior (ML)	prediction_pipeline.py: QuantilePredictor.predict_quantiles()	p10/p50/p90 per biomarker from the profile.	Every user needs a starting band, even with no data at all.
3. Wearable windows	wearable_windows.py: feature_windows()	30/90-day mean, SD, slope and coverage per metric; under 70% coverage = insufficient.	Turns noisy daily data into stable trends and handles forgotten days.
4. Usable labs	lab_anchor.py: anchoring_labs()	Keeps measured labs up to as_of, minus those cancelled by statin, BP medication or pregnancy.	A lab only anchors while it still describes the person.
5. Choose the band	twin_pipeline.py: _biomarker_prediction() using registry.lab_active() and lab_anchor.py: lab_band()	Lab band if a usable lab exists and the lab stage is on; otherwise the population band.	Never worse: the most personal proven layer wins.
6. Change flag	lab_anchor.py: change_flag()	With 2+ labs, compares the last two against the RCV.	Tells real change from lab noise.
7. Wearable shadow	twin_pipeline.py: _shadow(), residual_model.py: predict_band(); stored by shadow_store.py after the response	If a trained residual model exists and the user is eligible, computes a wearable-adjusted band; served only if registry.wearable_active().	Lets the wearable model be tested on real users without affecting them.
8. Risk	twin_pipeline.py: _risk(), _bmi() → risk.py: assess_risk() → prevent.py: prevent_risk()	Measured labs or predicted p50 go to the official AHA PREVENT package; corners give the range; BMI uses smart-scale weight from the last 30 days if present.	Industry-standard risk equations, not our own.
9. Reminder and audit	twin_pipeline.py: _lab_prompt(), input_hash(), _model_versions()	Next lab due 90 days after the latest lab (now if none); fingerprint of input and model versions.	Prompts the most useful next action; makes every answer reproducible.

Scenario A: Info only: no lab, no wearables

Code path. Steps 1-2 and 8-9 do the work. Step 4 finds no labs, so step 5 serves the population band for every biomarker. Risk uses predicted values only, so it is an estimate with a wide range, and the first lab is due now.

Biomarker	p10 / p50 / p90	basis	basis_label
Non-HDL chol. (mg/dL)	92 / **155** / 212	population	Typical for your profile
HDL chol. (mg/dL)	35 / **46** / 60	population	Typical for your profile
HbA1c (%)	5.3 / **5.8** / 6.6	population	Typical for your profile
eGFR (kidney)	81 / **101** / 109	population	Typical for your profile
Systolic BP (mmHg)	107 / **120** / 137	population	Typical for your profile

Risk: 3.3% (range 2.1-12.2), low, risk_basis *estimated*. **lab_prompt:** due now. **Wearables:** no wearable data. **Shadow:** none written (no trained residual model yet).

Scenario B: Lab from 2 days ago, no wearables

Code path. Step 1 converts the panel and derives non-HDL (230 − 38 = 192) and eGFR from creatinine. Step 4 keeps all five labs; step 5 serves lab bands centred on them. Risk now uses measured values, so it has no range, and rises to borderline because his real values are worse than typical. Next lab due 90 days after the lab.

Biomarker	p10 / p50 / p90	basis	basis_label
Non-HDL chol. (mg/dL)	165 / **192** / 224	lab	Based on your Nov 2 lab
HDL chol. (mg/dL)	34 / **38** / 43	lab	Based on your Nov 2 lab
HbA1c (%)	5.8 / **5.9** / 6.0	lab	Based on your Nov 2 lab
eGFR (kidney)	81 / **91** / 101	lab	Based on your Nov 2 lab
Systolic BP (mmHg)	124 / **138** / 154	lab	Based on your Nov 2 lab

Risk: 5.3%, borderline, risk_basis *measured*. **lab_prompt:** due 2027-01-31. **Wearables:** no wearable data. **Shadow:** none written (no trained residual model yet).

Scenario C: Lab from 45 days ago, no wearables

Code path. Same path as B. The only difference is step 5's drift term: 45 days of drift make the bands very slightly wider, and the next lab is due sooner.

Biomarker	p10 / p50 / p90	basis	basis_label
Non-HDL chol. (mg/dL)	164 / **192** / 225	lab	Based on your Sep 20 lab
HDL chol. (mg/dL)	34 / **38** / 43	lab	Based on your Sep 20 lab
HbA1c (%)	5.7 / **5.9** / 6.1	lab	Based on your Sep 20 lab
eGFR (kidney)	81 / **91** / 101	lab	Based on your Sep 20 lab
Systolic BP (mmHg)	124 / **138** / 154	lab	Based on your Sep 20 lab

Risk: 5.3%, borderline, risk_basis *measured*. **lab_prompt:** due 2026-12-19. **Wearables:** no wearable data. **Shadow:** none written (no trained residual model yet).

Scenario D: Wearables for 20 days, no lab

Code path. Step 3 runs, but 20 days cover only 57% of the 30-day window, under the 70% minimum, so the window is insufficient. Step 7 is not eligible. The result is identical to A.

Biomarker	p10 / p50 / p90	basis	basis_label
Non-HDL chol. (mg/dL)	92 / **155** / 212	population	Typical for your profile
HDL chol. (mg/dL)	35 / **46** / 60	population	Typical for your profile
HbA1c (%)	5.3 / **5.8** / 6.6	population	Typical for your profile
eGFR (kidney)	81 / **101** / 109	population	Typical for your profile
Systolic BP (mmHg)	107 / **120** / 137	population	Typical for your profile

Risk: 3.3% (range 2.1-12.2), low, risk_basis *estimated*. **lab_prompt:** due now. **Wearables:** 30-day window coverage 57%, insufficient. **Shadow:** none written (no trained residual model yet).

Scenario E: Wearables for 45 days, no lab

Code path. Step 3 now finds a sufficient 30-day window and returns it in wearable_windows. Step 7 would run in “level” mode, but no residual model has been trained, and the registry has the wearable stage off. Biomarker bands and risk are therefore still identical to A.

Biomarker	p10 / p50 / p90	basis	basis_label
Non-HDL chol. (mg/dL)	92 / **155** / 212	population	Typical for your profile
HDL chol. (mg/dL)	35 / **46** / 60	population	Typical for your profile
HbA1c (%)	5.3 / **5.8** / 6.6	population	Typical for your profile
eGFR (kidney)	81 / **101** / 109	population	Typical for your profile
Systolic BP (mmHg)	107 / **120** / 137	population	Typical for your profile

Risk: 3.3% (range 2.1-12.2), low, risk_basis *estimated*. **lab_prompt:** due now. **Wearables:** 30-day window coverage 90%, sufficient. **Shadow:** none written (no trained residual model yet).

Scenario F: Lab 75 days ago, then 70 days of wearables

Code path. Same lab path as B and C (step 5 serves lab bands). Steps 3 and 7: both windows are sufficient and there are more than 60 wearable days after the lab, so a trained “change”-mode model would write a shadow band to ml.shadow_predictions. The user would still see the lab band until the gate passes.

Biomarker	p10 / p50 / p90	basis	basis_label
Non-HDL chol. (mg/dL)	164 / **192** / 225	lab	Based on your Aug 21 lab
HDL chol. (mg/dL)	34 / **38** / 43	lab	Based on your Aug 21 lab
HbA1c (%)	5.7 / **5.9** / 6.1	lab	Based on your Aug 21 lab
eGFR (kidney)	81 / **91** / 101	lab	Based on your Aug 21 lab
Systolic BP (mmHg)	123 / **138** / 154	lab	Based on your Aug 21 lab

Risk: 5.3%, borderline, risk_basis *measured*. **lab_prompt:** due 2026-11-19. **Wearables:** 30-day window coverage 90%, sufficient. **Shadow:** none written (no trained residual model yet).

> **Takeaway for the backend:** today, wearables change only the `wearable_windows` field (good for an activity-trends card) and BMI when a smart-scale weight from the last 30 days exists. They do not change biomarker bands or risk until a residual model is trained on real paired data and passes the gate.

How the dashboard should show each case

B and C (recent or older lab): show the lab value as the headline, with its date; show the band as a secondary line. The lab anchor's p50 is the lab value itself, so there is no conflict between “lab” and “prediction”. What the anchor adds is the p10-p90 range, which means *where the next test will most likely land*, not a range for today's value. Never present it

as “your HDL is 34-43”: we measured 38.

Case	Headline	Second line	Extra
B: lab 2 days ago	HDL 38 mg/dL	Measured Nov 2 (2 days ago). Next test likely 34-43.	Source from basis_label.
C: lab 45 days ago	HDL 38 mg/dL	Measured Sep 20 (45 days ago). Next test likely 34-43.	Show the age clearly; “next lab by Dec 19” from lab_prompt.
A, D, E: no lab	HDL: likely 35-60 mg/dL	Typical for your profile.	Lead with the range, not p50; button: “upload a lab to see your value”.
F: lab + wearables	As B/C	As B/C	Activity-trends card from wearable_windows.
Lab past due	Keep the value	Dim it or highlight the date	“Time to retest” when lab_prompt.due is true.
2+ labs	Latest value	As B/C	change badge: “−22%, real change” or “+13%, within normal variation”.
Future: wearable layer on	Keep the measured lab	“Estimated today: 5.8 (from your activity)”	Shown when basis = lab_wearable; never replace a measured value with an estimate.

10. What the user experiences in the app

The engine returns labels and reasons so the app can explain itself in plain words. A suggested way to present each moment:

Moment	What the user might see	What is happening behind it
Signs up	“Based on people like you, your HDL is likely 35-60. Upload a lab to make this yours.” Risk shown as an estimate with a range.	Population prior (ML) only; risk_basis = estimated; lab prompt due.
Uploads a lab	“Your HDL is 38. Expect your next test between 34 and 43.” Risk now “based on your labs”.	Lab anchor replaces the prior; PREVENT uses measured values.
Rejected upload	“We could not use 1 record: unit not recognised.”	Ingestion rejected it with a reason; nothing else is affected.
Wears the watch	Trends (resting heart rate down, HRV up) shown as activity insights. Health numbers do not jump around day to day.	Windows computed; wearable model in shadow mode only.
Forgets the watch	Nothing breaks. If too many days are missing, trends are shown as “not enough data this month”.	Window below 70% coverage is left out.
90 days later	“Your last lab is 3 months old. A new test will sharpen your picture.”	lab_prompt.due = true.
Second lab	“Non-HDL down 22%: a real improvement.” / “HDL up 13%: promising, not yet beyond normal variation.”	RCV real-change test on the two labs.
Starts a statin	“Your cholesterol picture has been reset because your treatment changed. Please retest in 6-12 weeks.”	Lipid labs before the statin are ignored; population band returns.
Age under 30 or over 79	Biomarker bands shown; risk shown as not available.	PREVENT is validated only for ages 30-79.

Good practice for the app: always show the band, not just the middle value, and always show the basis label (from population / from your lab). Users trust a number more when they can see why it is what it is.

11. How we evaluate the output: steps and metrics

This section explains how each part of the engine is tested, which scores we look at, what each score means in plain words, and the actual results. All prior-model numbers below are from people the model never saw during training.

11.1 Which metrics fit which question

There is no single “accuracy” for this engine, because it answers different kinds of questions:

Question the engine answers	Type of problem	Metrics that fit
“What is this person's HDL likely to be?” (the middle value, p50)	**Regression**: predicting a number	MAE, RMSE, MSE, MAPE, R^2
“How sure are we?” (the band, p10 to p90)	**Interval / probabilistic** prediction	Coverage, band width, pinball loss, skill score
“Is the band equally honest for everyone?”	**Fairness**	Coverage gap between sexes and age groups
“Did this lab really change?” (RCV flag: yes/no)	**Classification**	False-alarm rate (specificity), recall (detection rate)
“Does this person likely have diabetes-range HbA1c?” (yes/no)	**Classification**	Accuracy, precision, recall, specificity

> **Why not accuracy, precision and recall everywhere?** They need yes/no answers. Predicting an HDL of 47 when the truth is 48 is not “wrong”, just 1 mg/dL off, so number-predictions are scored by how far off they are (MAE, RMSE) instead. We use accuracy, precision and recall only where the engine really makes a yes/no call.

11.2 What each metric means

Metric	Formula (plain)	What it tells you	Good value
MAE (mean absolute error)	average of |truth − prediction|	Typical miss, in the biomarker's own unit. Easiest to explain.	Lower
MSE (mean squared error)	average of (truth − prediction)²	Like MAE but big misses count much more. Unit is squared, so hard to read.	Lower
RMSE (root MSE)	√MSE	MSE back in the biomarker's unit. Always ≥ MAE; a big gap between them means some large misses.	Lower
MAPE	average of |error| / truth, in %	Typical miss as a percentage, so biomarkers can be compared.	Lower
R^2 (R-squared)	1 − MSE / variance of truth	Share of the differences between people that the model explains. 0 = no better than guessing the average; 1 = perfect.	Higher
Coverage	share of truths inside p10 to p90	Is the 80% band honest?	≈ 80%
Band width	average of p90 − p10	How precise the band is. Only meaningful once coverage is right.	Narrower
Pinball loss	penalty that is asymmetric around each percentile	The correct score for a p10, p50 or p90 prediction (what the model is trained on).	Lower
Skill score	1 − model loss / baseline loss	Improvement over a naive baseline (guess everyone the same value). 0.2 = 20% better.	Higher

Metric	Formula (plain)	What it tells you	Good value
Accuracy	correct yes/no calls / all calls	Overall share right. Misleading when “yes” is rare.	Higher
Precision	true yes / all predicted yes	When we say yes, how often we are right.	Higher
Recall (sensitivity)	true yes / all real yes	Of the real cases, how many we catch.	Higher
Specificity	true no / all real no	Of the non-cases, how many we correctly leave alone. 1 − specificity = false-alarm rate.	Higher

All averages are **survey-weighted**: each NHANES person counts as many times as the number of Americans they represent, so results describe the US adult population.

11.3 The evaluation steps

Step	What we do
1. Hold out unseen people	20% of NHANES people are set aside before training (grouped by survey cluster so neighbours do not leak into the test). Every prior-model score is computed on them.
2. Tune with grouped 5-fold cross-validation	On the other 80%, settings (tree size, number of trees) are chosen by rotating training and testing across 5 groups of clusters, scored by pinball loss.
3. Compare with a naive baseline	Baseline = give everyone the same population percentile. The model's pinball loss is compared with it (skill score) and its MAE with the baseline's MAE.
4. Score the middle value (p50)	MAE, MSE, RMSE, MAPE, R^2 on the held-out people.
5. Score the band	Coverage of the 80% band and its average width.
6. Check fairness	Coverage per sex and per age band; the largest gap must be small. Measured with cross-conformal evaluation (every person scored once while out of training) for stable numbers.
7. Score the lab anchor	Simulated repeat labs (section 11.5): error vs the population band, coverage, width, false alarms. This decides whether the lab layer is switched on.
8. Score the yes/no calls	Real-change flag: false alarms and detection rate. Diabetes-range HbA1c from the prior: accuracy, precision, recall, specificity.
9. Gate the wearable model	On real users only: pinball loss 10% better than both lower layers, bootstrap lower bound above 0, coverage 75-85%, at least 200 users (section 7).
10. Check the service	Response time, and that the same input always gives the same output (input_hash).

11.4 Results: population prior (ML), held-out people

Biomarker	MAE	RMSE	MSE	MAPE	R^2	Baseline MAE	Coverage	Width
Non-HDL (mg/dL)	26.1	34.8	1213.6	21.6%	0.23	30.5	82%	91.0
HDL (mg/dL)	8.8	12.0	143.7	16.7%	0.26	10.6	82%	30.0
HbA1c (%)	0.40	0.75	0.56	6.4%	0.47	0.55	82%	1.19
eGFR	12.5	15.7	247.6	16.8%	0.41	16.8	80%	35.5
Systolic BP (mmHg)	11.3	15.0	225.9	9.3%	0.23	13.3	79%	34.7

Biomarker	Skill p10	Skill p50	Skill p90	Largest subgroup coverage gap
Non-HDL (mg/dL)	7%	14%	13%	0.5 points
HDL (mg/dL)	8%	17%	17%	0.2 points
HbA1c (%)	14%	27%	51%	0.7 points
eGFR	23%	25%	40%	0.1 points
Systolic BP (mmHg)	9%	16%	14%	0.1 points

How to read this. The model beats the naive baseline for every biomarker (lower MAE, positive skill), and its 80% bands are honest (79-82% coverage, subgroup gaps under 1 point). But R^2 is modest (0.23-0.47): age, sex, weight and history explain only part of why one person's cholesterol differs from another's. That is expected, and it is exactly why the engine replaces the prior with the user's own lab as soon as one exists. The prior's job is an honest starting band, not a precise personal value.

11.5 How the lab anchor is evaluated, step by step

The question is simple: **when a user has a lab, does the lab band predict their next lab better than the population band does?** Answering it needs people with two labs some months apart. NHANES has only one visit per person, and our users do not have repeat labs yet, so the evaluation **simulates** the second lab. The code is `simulate_lab_stage()` in `simulation.py`, run by `evaluate_lab_stage.py`; it writes the results and the on/off decision into the activation registry.

Step	What is done	Why
1. Take unseen people	The same held-out 20% of NHANES used for the prior (never used in training), one biomarker at a time. Each person's measured value is treated as their true level.	Honest test: the population band is not scored on people it learned from.
2. Simulate their first lab	first lab = true level × random lab noise (published CV, F3).	A real lab is never exactly the true level.
3. Pick when the next lab happens	A random gap of 90 to 365 days per person.	Covers the typical range of follow-up intervals.

Step	What is done	Why
4. Simulate drift and the next lab	next true level = true level × random drift for that gap (F4); next lab = next true level × new random lab noise.	The thing we try to predict is a real-world next lab: changed a little, measured noisily.
5. Build both bands	Lab band: the engine's formulas around the first lab (F5, F6). Population band: the prior's p10/p50/p90 for each person's profile.	Compares the two layers the registry chooses between.
6. Score accuracy	Median absolute error = typical |next lab − band centre|, survey-weighted, for each band. Error cut = 1 − lab error / population error.	Median, not mean, so a few extreme simulated values do not dominate.
7. Score honesty	Coverage = share of next labs inside each 80% band.	A band is only useful if it is honest (target 75-85%).
8. Score precision	Width ratio = average lab-band width / average population-band width.	Below 1 means the lab band is narrower.
9. Score the change flag	A second, stable repeat with no drift is simulated; the share flagged as “real change” by the RCV is the false-alarm rate.	Nothing changed, so every flag is a false alarm (target about 5%).
10. Decide on or off	The lab layer is switched on only if: lab error < population error, AND width ratio < 1, AND lab coverage 75-85%. The decision and all metrics are saved in config/activation.json.	The “never worse” rule, applied automatically per biomarker.

Limitation to keep in mind. The simulation draws noise using the same published CVs that the lab band uses. So it proves the lab band is *consistent* with that noise and that it beats the population band, but it cannot prove the CVs themselves are right for our users. The fix is to re-run steps 6-10 on real users' repeat labs (first lab → band, next real lab → score) as soon as enough exist. The metrics and pass rules stay the same. A fixed random seed (2026) makes every run reproducible.

11.6 Results: lab anchor vs population prior

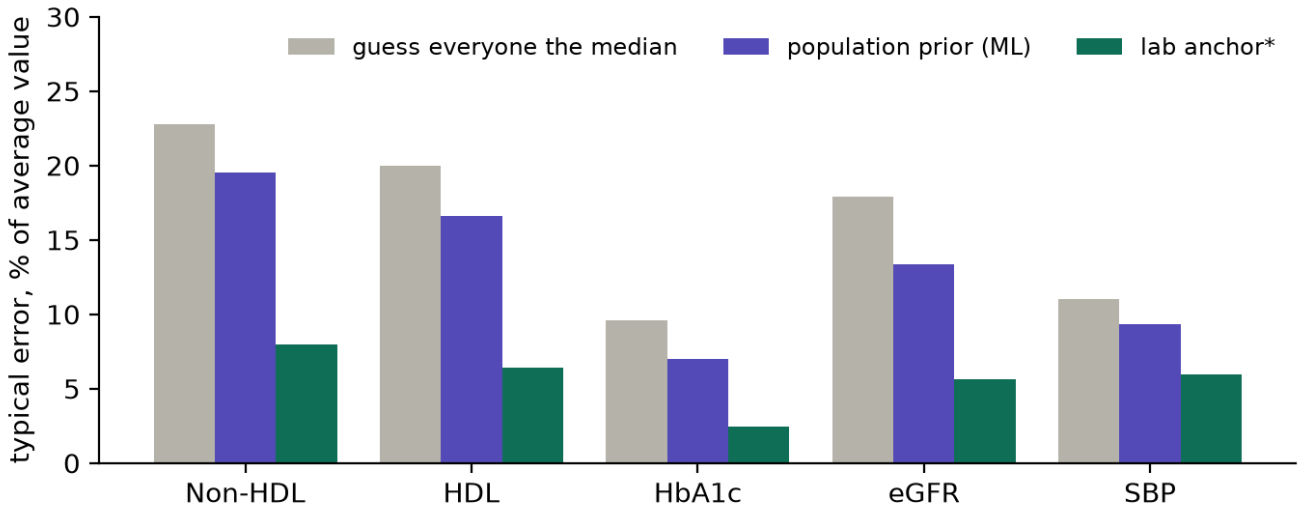


Figure 14. Typical error as % of the average value. Grey and purple: mean absolute error on held-out NHANES people. *Green: median absolute error on simulated next labs (NHANES has no repeat labs), so it is indicative, not a like-for-like comparison.

Biomarker	Lab median abs. error	Population median abs. error	Error cut	Lab coverage	Width vs population
Non-HDL (mg/dL)	10.6	20.9	49%	82%	52%
HDL (mg/dL)	3.4	6.9	51%	82%	47%
HbA1c (%)	0.14	0.25	45%	79%	45%
eGFR	5.3	10.1	48%	81%	60%
Systolic BP (mmHg)	7.2	9.7	26%	80%	83%

11.7 Results: yes/no decisions

Real-change flag (RCV). Two kinds of mistakes matter: flagging noise as a real change (false alarm), and missing a real change. We simulated 200,000 pairs of labs per biomarker using the published noise:

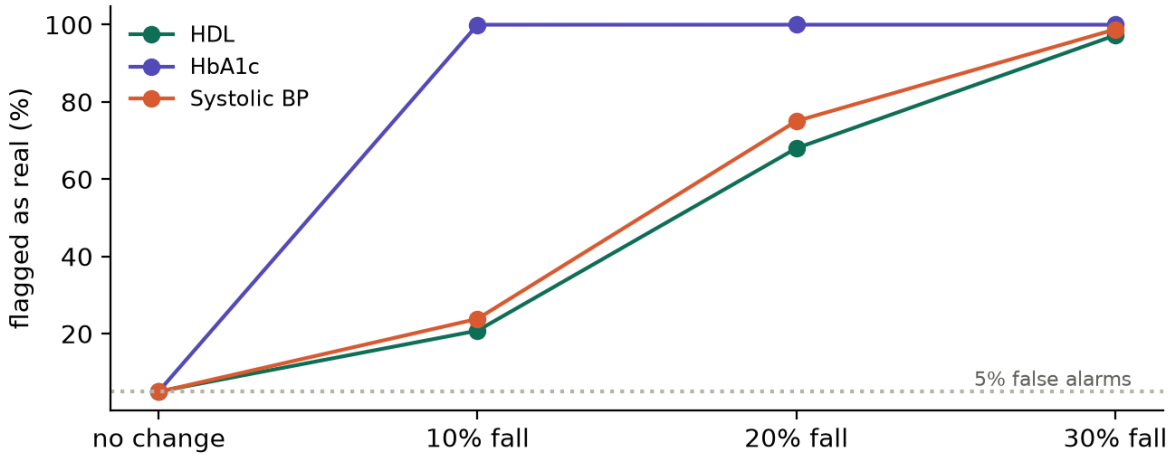


Figure 15. How often a change is flagged as real. With no true change, about 5% are flagged (false alarms, by design). Larger true changes are caught more often. HbA1c is measured so precisely that even a 10% fall is caught most of the time.

Biomarker	False alarms (no change)	Recall: 10% fall	Recall: 20% fall	Recall: 30% fall
Non-HDL (mg/dL)	5%	14%	46%	84%
HDL (mg/dL)	5%	21%	68%	97%
HbA1c (%)	5%	100%	100%	100%
eGFR	5%	24%	75%	99%
Systolic BP (mmHg)	5%	24%	75%	99%

Precision is not reported for this flag, because it depends on how often real changes happen among our users, which we do not know yet. It can be measured once users have repeat labs.

Diabetes-range HbA1c from the population prior. For a user with no HbA1c lab, does the prior's typical value (p50) say 6.5% or higher? Scored on held-out people, where 10% truly are in that range:

Accuracy	Precision	Recall	Specificity	Recall using the high end (p90 ≥ 6.5%)
95%	73%	79%	97%	88%

Accuracy looks high partly because most people are not in the diabetes range. Precision and recall are the more honest view: when the prior suggests diabetes-range HbA1c it is right about 3 times in 4, and it catches about 4 in 5 real cases. That is why an HbA1c lab is prompted early: the prior is a screening hint, not a diagnosis. (The engine itself treats diabetes as a PREVENT input only from a self-report or a measured HbA1c of 6.5% or more.)

11.8 What we do not evaluate ourselves

- **PREVENT risk accuracy.** We run the AHA's equations unchanged. Their discrimination (how well they rank who has an event) and calibration (whether predicted risk matches observed

rates) were evaluated by the AHA in millions of people [see section 13, reference 8]. We check only that our inputs are correct and that results match the official package.

- **The wearable layer.** Not evaluated yet: it needs real users with wearables followed by a lab. The gate in section 7 is the evaluation plan.
- **Real repeat labs.** The lab-anchor results are simulated. Once users have repeat labs, steps 7 and 8 will be re-run on real data with the same metrics.

12. How we know it works, and its honest limits

Testing the lab anchor

NHANES has only one visit per person, so true repeat labs are not available yet. We therefore **simulated** them: for each held-out person we generated a first lab and a later lab 90-365 days apart, using the same published noise and drift, and scored both the lab band and the population band against that later lab.

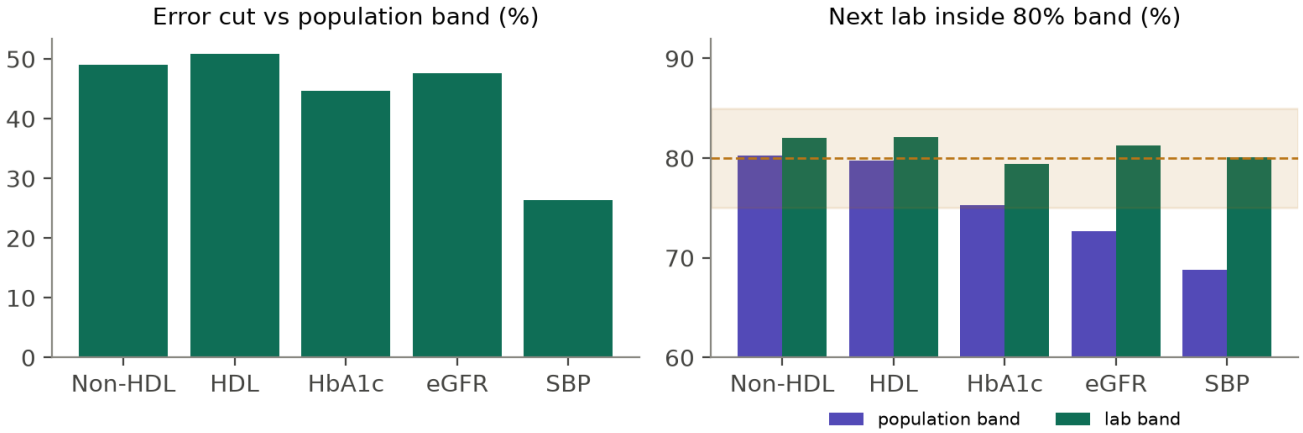


Figure 16. Left: how much the lab anchor cut the typical error versus the population guess. Right: how often the next lab landed inside the 80% band (target zone 75-85% shaded). The population band falls short for HbA1c, eGFR and SBP here because the simulated next lab adds measurement noise and drift that the population band was not built for.

Goal	Result	Met?
Error cut of at least 40% vs population	Non-HDL 49%, HDL 51%, eGFR 48%, HbA1c 45%; SBP 26%	4 of 5 biomarkers
80% band covers 75-85% of next labs	79-82% for every biomarker	Yes
Band at most one-third of population width	45-83% of population width	No: lab noise alone makes this impossible
Subgroups within a few points of each other	Largest gap 0.7 points	Yes
False “real change” alarms about 5%	4.3-5.7%	Yes (by design)
Wearable layer beats lab and population	Waiting for real paired data	Not testable yet
Fast enough for the app	95% of requests answered in under 42 ms	Yes

Honest limits

- The lab-anchor tests are simulations built from published noise values, not real repeat labs. They show the design is consistent; real repeat labs from users must confirm it.
- Blood pressure did not reach the 40% error-cut target: home readings are noisy and the population model already predicts it fairly well, so a lab adds less.
- Without labs, the engine can describe what is typical for someone like you, not your true value. The app should say so.
- The wearable layer is not proven. Until it passes the gate, wearables do not change anyone's health numbers or risk.
- PREVENT estimates population risk; it is a conversation starter with a clinician, not a diagnosis.

13. Sources: every published number and where it comes from

Every fixed number in the engine falls into one of four kinds. This section lists each one, what it is used for, and its source, so anyone can check it.

Kind	Meaning	Can we change it?
Physical constant	Follows from chemistry or a legal definition (e.g. a molecule's weight).	No: it is a fact.
Published standard	Taken from a peer-reviewed paper, guideline or official package.	Only if the source is updated.
Published estimate	Measured by research studies; we picked a value from their range.	Yes, if better evidence appears (including our own data).
Our design choice	Set by us, not taken from research. Tested where possible.	Yes, it is a product decision.

References are numbered [1], [2], ... and listed at the end of this section.

13.1 Unit conversions (record ingestion)

Labs and devices report values in different units. Ingestion converts everything to one unit per biomarker before any statistics run. These are the constants in `src/components/biomarkers.py`.

Constant	Used for	Where it comes from	Kind
MG_DL_PER_MMOL_L_CHOLESTEROL = 38.67	Total, HDL, non-HDL and LDL cholesterol reported in mmol/L (common outside the US) → mg/dL. Example: HDL 1.34 mmol/L × 38.67 = 52 mg/dL.	Cholesterol's molar mass is 386.65 g/mol [1]. 1 mmol/L = 386.65 mg/L = 38.665 mg/dL, rounded to 38.67. The same factor is used for all cholesterol fractions because they are all measured as cholesterol.	Physical constant
UMOL_L_PER_MG_DL_CREATININE = 88.42	Creatinine reported in µmol/L → mg/dL (divide by 88.42), before eGFR is computed. Example: 88 µmol/L ÷ 88.42 = 1.0 mg/dL.	Creatinine's molar mass is 113.12 g/mol [2]. 1 mg/dL = 10 mg/L ÷ 113.12 g/mol = 88.4 µmol/L.	Physical constant
KG_PER_LB = 0.45359237	Smart-scale weight in pounds → kg (for BMI and the weight wearable metric).	Exact by definition: the international pound has been defined as exactly 0.45359237 kg since 1959 [3].	Physical constant (exact)

Constant	Used for	Where it comes from	Kind
HBA1C_NGSP_SLOPE = 0.09148 HBA1C_NGSP_INTERCEPT = 2.152	HbA1c reported in mmol/mol (IFCC units, used in Europe) → % (NGSP units, used in the US and by PREVENT): % = 0.09148 × mmol/mol + 2.152. Example: 48 mmol/mol → 6.5%.	The IFCC-NGSP “master equation” from a method-comparison study across the US, Japanese and Swedish standardization networks [4], published by the NGSP [5].	Published standard

13.2 Derived values (computed from other labs)

Value / constants	Used for	Where it comes from	Kind
Non-HDL = total cholesterol − HDL	Builds non-HDL when a lab reports total and HDL on the same day.	Standard definition used in cholesterol guidelines [12].	Published standard
eGFR, CKD-EPI 2021: 142, κ 0.7 (F) / 0.9 (M), α −0.241 (F) / −0.302 (M), −1.200, 0.9938^{age}, ×1.012 (F)	Kidney function from creatinine, age and sex; one of the five predicted biomarkers and a PREVENT input.	Race-free CKD-EPI creatinine equation [6]. Chosen because PREVENT was built with this same equation [8].	Published standard
BMI = weight (kg) / height $(m)^2$	PREVENT input.	Standard WHO/CDC definition.	Published standard

13.3 Lab noise and drift (lab anchor)

Stored in `src/components/personalization/biomarker_config.py`. This is where the CV of a single lab comes from (section 5b).

Biomarker	measurement_cv	annual_drift_cv
Non-HDL cholesterol	8.5%	8%
HDL cholesterol	6.5%	6%
HbA1c	1.3%	4%
eGFR (from creatinine)	6.0%	4%
Systolic BP (home protocol)	6.0%	5%

measurement_cv: how each number is derived. Within-person biological CVs (CV_I) are the EFLM meta-analysis medians for serum/plasma (whole blood for HbA1c), database update of 10 August 2026, read on 4 November 2026 [7]. The analytical CV is the EFLM desirable specification $CV_A = 0.5 \times CV_I$ [23], an assumption until our partner lab's quality-control CV replaces it. Total = $\sqrt{(CV_I^2 + CV_A^2)}$ (F2).

Biomarker	Published input (95% CI)	Derivation	Evidence-based total	Engine uses	Verdict
HDL cholesterol	CV_I 5.7% (4.7-7.8) [7]	$\sqrt{(5.7^2 + 2.85^2)}$	6.4%	6.5%	Matches
Non-HDL cholesterol	No EFLM entry. Total chol. CV_I 5.4% (4.9-6.6); LDL CV_I 7.6% (7.0-8.4) [7]	Non-HDL = total − HDL: $SD^2 = SD_{total}^2 + SD_{HDL}^2$ at NHANES means (187 and 53 mg/dL) → 8.8% (upper bound if the two are positively correlated). LDL as a proxy: 8.5%.	about 8.5%	8.5%	Matches (updated from 7.0%)
HbA1c (NGSP %)	CV_I 1.2% (0.2-7.0); IFCC units 1.6% (1.3-2.5) [7]	$\sqrt{(1.2^2 + 0.6^2)}$	1.3%	1.3%	Matches (updated from 2.5%)
eGFR (from creatinine)	Creatinine CV_I 4.4% (2.3-4.7) [7]	Creatinine total $\sqrt{(4.4^2 + 2.2^2)} = 4.9\%$; eGFR ∝ creatinine$^{-1.2}$ [6], so CV ≈ 1.2 × 4.9%	5.9%	6.0%	Matches
Systolic BP (home)	Day-to-day CV of one morning reading 5.4 ± 2.4% (evening 6.1%) in 368 CKD patients [25]; top 10% of a general population above 11.0% [26]	Single reading, taken as is	5-6%	6.0%	Matches a single reading

annual_drift_cv: what the evidence says.

Biomarker	Engine uses	Evidence	Verdict
Non-HDL / HDL	8% / 6%	In 9,014 LIPID-trial patients, within-person SD of total cholesterol rose from 7% to 11% and long-term variation took almost 4 years to equal the short-term 7% [24], i.e. about 7% / √4 = 3.5% per year.	Wider than the evidence (cautious)
HbA1c, eGFR, SBP	4%, 4%, 5%	No published source found.	**Our design choice**

What this means. All five noise values now follow the published data. Two were corrected on 4 November 2026 after this check: **non-HDL from 7.0% to 8.5%** (it was too narrow, so next labs would have fallen outside the 80% band too often) and **HbA1c from 2.5% to 1.3%** (it was too wide, so the real-change flag missed smaller real changes; its limit is now about ±3.6% instead of ±7%). The lab-stage evaluation was re-run and all five lab layers remain on. The analytical part still assumes the EFLM desirable level and should be replaced with our partner lab's quality-control CV. Drift values for HbA1c, eGFR and blood pressure have no source and should be replaced with estimates from our users' repeat labs once enough exist.

13.4 Statistical constants and methods

Number / method	Used for	Where it comes from	Kind
1.2816	Turns a spread into an 80% band (p10, p90).	Standard normal quantile: 10% of a normal distribution lies below −1.2816 SD.	Mathematical constant
1.96	95% real-change limit (RCV).	Standard normal quantile: 2.5% lies beyond ±1.96 SD on each side.	Mathematical constant
$\sigma = \sqrt{\ln(1 + CV^2)}$	CV → log-scale spread.	Exact property of the log-normal distribution [9].	Mathematical identity
Reference Change Value	Real change vs noise between two labs.	RCV concept [10]; log-normal form with different fall/rise limits [11].	Published standard
Drift grows with √(time)	Band widening after a lab.	Random-walk assumption.	Our design choice
Theil-Sen slope	Wearable trend per window.	Theil [17]; Sen [18].	Published method

13.5 Risk calculation (PREVENT)

Number / rule	Used for	Where it comes from	Kind
All risk coefficients	10- and 30-year risk of total CVD, ASCVD, heart failure.	AHA PREVENT equations [8], run unchanged through the official AHAprevent R package version 1.1.0 [13]. We hold no coefficients of our own.	Published standard
Age 30-79; total cholesterol 130-320 mg/dL; HDL 20-100; SBP 90-200 mmHg; BMI 18.5-39.9	Inputs outside are clipped and flagged; risk is not generated outside ages 30-79.	Validated ranges stated in the AHA package's own code (aha_prevent_base.ado) [13] and the PREVENT paper [8].	Published standard
30-year risk only for ages 30-59	Whether 30-year risk is returned.	PREVENT paper [8].	Published standard
HbA1c ≥ 6.5% counts as diabetes	PREVENT diabetes input when a lab shows it, even if not self-reported.	ADA diagnostic threshold [14].	Published standard
Categories: low <5%, borderline 5-7.5%, intermediate 7.5-20%, high ≥20%	Label shown next to 10-year total CVD risk.	Thresholds from the ACC/AHA cholesterol and primary-prevention guidelines [12, 15], which defined them for the older Pooled Cohort Equations. Applying them to PREVENT total CVD is **our choice**; PREVENT estimates tend to be lower, so this should be confirmed with clinical advisors.	Published standard, adapted

13.6 Data and machine-learning methods

Item	Used for	Source
NHANES August 2021-August 2023	Training data for the population prior, with survey weights and design clusters.	CDC National Center for Health Statistics [16].
LightGBM	Population prior and wearable residual model.	Ke et al. [19].
Quantile regression / pinball loss	Training p10, p50, p90.	Koenker & Bassett [20].
Conformalized quantile regression	Widening bands so they truly cover 80%.	Romano, Patterson & Candès [21]; per-group (Mondrian) version: Vovk et al. [22].

13.7 Our own design choices (not from research)

These are product decisions. They are listed so nobody mistakes them for published facts.

Setting	Value	Why
Band level	80% (p10 to p90)	Wide enough to be honest, narrow enough to be useful; same for every layer.
Plausible ranges	e.g. HDL 5-200 mg/dL	Reject typing and unit errors; also used as the training outlier rule.
Wearable windows	30 and 90 days	Short-term and longer-term trend.
Minimum window coverage	70% of days	Enough days for a stable mean and trend, while tolerating forgotten devices.
Smart-scale weight freshness	30 days	Older weights no longer describe the user.
Next-lab reminder	90 days after the latest lab	Typical follow-up interval.
Wearable gate	200 users, 10% better, coverage 75-85%, bootstrap lower bound above 0	Enough evidence before changing what users see.
Activation order	HbA1c, then HDL and non-HDL; eGFR never	Most plausible links between daily habits and the biomarker first.
Minimum calibration group	150 people	Avoid widening a band from too few examples.

13.8 References

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14. Full glossary

Term	Meaning
AHA PREVENT	American Heart Association equations (2023) giving 10- and 30-year risk of heart disease, heart attack/stroke (ASCVD) and heart failure.
Activation registry	Settings file saying which layer is switched on for each biomarker, with the evidence.
Anchor reset	Dropping old labs after a statin, blood-pressure drug or pregnancy.
Basis	Which layer produced a band: population, lab, population_wearable, lab_wearable.
Bootstrap	Re-running a test many times on re-sampled users to see how stable the result is.
Calibration	Making a model's stated confidence match reality.
Conformal widening (Mondrian)	Stretching bands using test results so they really cover 80%, done separately per sex and age group.
Cross-validation (grouped)	Train on some data, test on unseen data, rotating; related people are kept in the same part.
CV (coefficient of variation)	Noise as a percentage of the value.
Decision tree	A flowchart of yes/no questions that ends in a number.
Drift	Slow, natural wandering of a person's true level over months.
eGFR	Estimated kidney filtering rate, computed from blood creatinine (CKD-EPI 2021 equation).
Gate	The set of tests the wearable model must pass before users see it.
Gradient boosting	Building many small trees, each fixing the previous ones' errors.
HbA1c	Average blood sugar over about 3 months, in %.
HDL / non-HDL cholesterol	“Good” cholesterol / all the other cholesterol (total minus HDL).
HRV	Heart-rate variability: beat-to-beat variation; higher usually means better recovery.
Ingestion	Checking, converting and cleaning incoming records.
input_hash	A fingerprint of the exact input and model versions, so any answer can be reproduced.
Lab anchor	Centring the band on the user's own latest lab, widened for known lab noise and time since the lab.
LightGBM	A fast, widely used gradient-boosting library.
Log scale	Working in percentages instead of raw units.
NHANES	US National Health and Nutrition Examination Survey.
p10 / p50 / p90	Low end, typical value and high end of the 80% band.

Term	Meaning
Pinball loss	The scoring rule that trains a model to hit a chosen percentile.
Population prior	The ML model's band for people with the same profile.
Quantile regression	Predicting percentiles (p10, p50, p90) instead of an average.
RCV (reference change value)	The largest change between two labs that noise alone explains 95% of the time.
Residual	The gap between what we predicted and what was measured.
Shadow mode	Running a model and saving its answers without showing them.
Survey weight	How many people in the population one NHANES participant represents.
Theil-Sen slope	A trend line that ignores a few strange days.
VO2 max	An estimate of aerobic fitness from the wearable.
Window / coverage	A 30- or 90-day summary / the share of its days that have data (70% needed).